1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Canonical direction is testable: PTEN loss in prostate cancer leads to constitutive activation of the PI3K/AKT/mTOR axis, as evidenced by elevated phosphorylated AKT (Ser473 and Thr308) and upregulated downstream signaling (p-GSK3β, p-S6K, p-4EBP1). This hyperactivation drives increased proliferation, enhanced survival, metabolic reprogramming (elevated glycolysis and lipid synthesis), and greater invasive potential. The degree of dependency on this pathway is modulated by PTEN gene dosage (homozygous loss vs haploinsufficiency) and by co-occurring alterations (ERG rearrangements, AR amplification, SPOP mutations). Therapeutically, PTEN-deficient prostate cancer cells will show superior sensitivity to PI3K/AKT/mTOR inhibitors compared with PTEN-competent cells, and combining AKT pathway blockade with AR pathway inhibition will produce synergistic anti-tumor effects, delaying progression to castration-resistant prostate cancer (CRPC). Additionally, compensatory activation of alternative survival pathways (e.g., MAPK/SGK) may arise upon PI3K/AKT inhibition and could be targeted with rational combination strategies to prevent resistance.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: are reliable readouts of pathway activation. PTEN status interacts with AR signaling and other genomic alterations, influencing tumor behavior and therapy responses.

Prediction 2: Generate isogenic PTEN-knockout and PTEN-wild-type prostate cancer cell lines (e.g., via CRISPR/Cas

Prediction 3: and patient-derived organoids with defined PTEN status.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions